Lightweight and Lifelong Hyperspectral Image Classification via Attention-Based Reservoir Computing

Anran Yuan, *Member, IEEE*, Dingchen Wang, Jing Bai, *Senior Member, IEEE*, Zhu Xiao, *Senior Member, IEEE*, Zhongrui Wang, *Member, IEEE*, Jianqing Li, *Senior Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— The continual progression and expanding applications of hyperspectral image (HSI) technology necessitate the development of lightweight HSI classification models that are capable of lifelong learning. However, the computationally-demanding task of training and updating HSI classification models, exacerbated by the substantial number of trainable parameters in feature extractors, remains a substantial challenge. This article proposes an attention-based reservoir computing (ARC) model to overcome these hurdles. The ARC model utilizes a cross-slicing operation to generate multidirectional inputs, treating the HSI dataset as a spatial sequence for processing within a reservoir. For every target pixel, four spatial sequences from various directions are introduced into the reservoir, generating four corresponding outputs. A voting mechanism then evaluates these outputs to yield the final prediction. In addition, we design an attention-based leaky function for reservoir computing (RC) to capture the spatial correlation inherent in HSI data accurately. The attention-based leaky function enables the reservoir state to weigh less on the pixels outside the region of interest (ROI) and have a longer memory for pixels inside the ROI. The ARC was tested on widely used HSI datasets: Indian Pines, PaviaU, and Salinas. It demonstrated competitive lightweight classification performance against state-of-the-art lightweight models by maintaining comparable training time while achieving superior accuracy. Furthermore, the model's lifelong learning accuracy also showed superior performance compared to existing lifelong learning models, with a one thousand times reduction of parameter-to-be-updated. This work makes the ARC model an effective contender for HSI classification tasks, excelling in both lightweight classification and lifelong learning capacities. The source codes are available publicly at: https://github.com/Waterman-Ann/ARC.**

***Index Terms*— Attention mechanism, hyperspectral image (HSI) classification, lifelong learning, lightweight network.**

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Anran Yuan is with the School of Artificial Intelligence, Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University, Xi'an 710071, China, and also with the School of Computer Science and Engineering, Macau University of Science and Technology, Macau, China (e-mail: annora.yuan@gmail.com).

Dingchen Wang and Zhongrui Wang are with the Department of Electrical and Electronic Engineering, The University of Hong Kong, Hong Kong (e-mail: dchwang@connect.hku.hk; zrwang@eee.hku.hk).

Jing Bai and Licheng Jiao are with the School of Artificial Intelligence, Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University, Xi'an 710071, China (e-mail: baijing@mail.xidian.edu.cn; lchjiao@mail.xidian.edu.cn).

Zhu Xiao is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China, and also with the Shenzhen Research Institute, Hunan University, Shenzhen 518055, China (e-mail: zhxiao@hnu.edu.cn).

Jianqing Li is with the School of Computer Science and Engineering, Macau University of Science and Technology, Macau, China (e-mail: jqli@must.edu.mo).

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I. Introduction

HYPERSPECTRAL image (HSI) technology has experienced rapid development in the last decade. The advancement in HSI technology brings numerous benefits to various downstream applications, such as natural resource management, environmental monitoring, and disaster warning [1], [2]. At the same time, the advancement in HSI technology brings challenges to the development of HSI classification models. On one hand, the expansion of HSI technology enables an increasing number of classification scenarios due to its broader imaging area [3]. On the other hand, the classification categories within each scenario undergo frequent updates as the revisiting periods become progressively shorter [4]. Consequently, there is a growing demand to train new classification models for emerging application scenarios and update existing models to accommodate incremental classes. However, the training of HSI classification models is a time-consuming process, which is often exacerbated by the large volume of trainable parameters in deep learning models. The training process for each new HSI classification model requires significant computational power and GPU time Fig. 1(a) and (c). To make things worse, the trained models are often required to be updated frequently to accommodate new classes Fig. 1(b). This is because the HSI classification models are often trained on a small number of labeled samples, which may not be sufficient to cover all the classes in the application scenario [5]. Updating of HSI model also incurs significant

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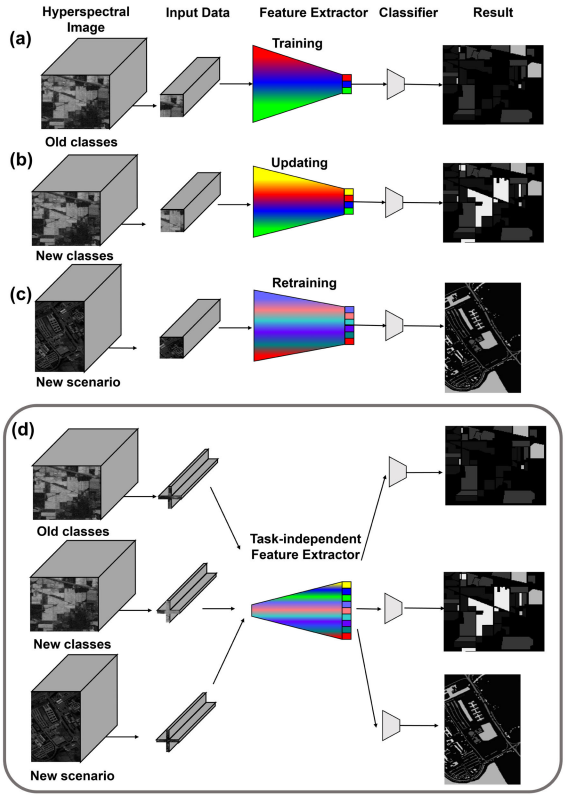


Fig. 1. Comparison of a typical HSI classification paradigm, an HSI incremental learning paradigm, and a lightweight classification paradigm with lifelong learning capacity. (a) Model training on old data class. (b) Model updating upon the emergence of the new class. (c) Model retraining upon the emergence of a new scenario. (d) Lightweight classification model with a task-independent feature extractor with the ability to adapt to new classes and scenarios in lifelong learning.

computational overhead, which is often not affordable in practice [2], [3].

To address the high computational costs associated with the training and updating of HSI classification models, efforts have been focused on developing lightweight models and incremental learning models. Lightweight models primarily aim to minimize the computational burden through model compression or parameter reduction techniques. Conversely, incremental learning models update the model to incorporate new classes without resorting to complete retraining. However, current models do not effectively integrate these two abilities.

Incremental learning models possess feature extractors with an extensive number of parameters, due to the complexity and high dimensionality of HSI data [6], [7], [8]. This characteristic makes the training process challenging and impedes the creation of lightweight classification models. In a similar vein, the feature extractors of lightweight models, which are trained in a task-specific manner, necessitate updates to facilitate new classes, thereby hindering its lifelong learning ability [9], [10]. To bridge this gap, it is required to address the following challenges.

1) The construction of a lifelong learning model with minimized training costs associated with complex feature extractors.
2) The development of a lightweight HSI classification model with a task-independent feature extractor that does not demand updates for new classes.

In light of these challenges, we propose a lightweight and lifelong HSI classification model, namely, attention-based reservoir computing (ARC). ARC eliminates the need for feature extractor training and updates. Specifically, we use cross-slicing operations without principal component analysis (PCA) at the spectral band to obtain the multidirectional spatial sequence with full spectral information, as shown in Fig. 1(d). For each target pixel, four spatial sequences coming from four directions are obtained. Each spatial sequence is then fed into a feature extractor inspired by the reservoir computing (RC) paradigm [11]. RC paradigms have been generally applied in various fields like speech recognition and robot control, due to their capability to nonlinearly encode sequential data into high-dimension separable feature space. The ARC encodes the spatial–spectral information into the reservoir state over the spatial steps at a certain leaky rate. To exploit the spatial correlation nature of HSI data, we propose an attention-based leaky function whose value is large at the region of interest (ROI) edge and decays inside the ROI. The leaky function allows the reservoir state to weigh less on spatial–spectral information outside the ROI and have a longer memory for spatial–spectral information inside the ROI. After the whole spatial sequence is fed into the reservoir, the reservoir state is the extracted feature, where a well-trained lightweight classifier is used to generate an output prediction. Then, the outputs from the four spatial sequences are voted to obtain the final prediction.

The ARC model is inherently lightweight due to the minimal number of trainable parameters. In ARC, training is only necessitated for the classifier's weights, while the reservoir's internal weights are randomly assigned and steadfast. This approach significantly simplifies the learning process, reducing both computational resources and time. In addition, ARC enables lifelong learning by continuously integrating new data through simple adjustments of the output weights, with all internal weights remaining fixed.

The article makes the following contributions.

1) We proposed the ARC model with a reservoir containing a randomly fixed internal connection, which serves as a task-independent feature extractor. The ARC is intrinsically lightweight by eliminating the need to train the feature extractor. With the nonlinear expression ability of the reservoir state, the ARC model outperforms other lightweight models that have a trainable feature extractor.
2) The proposed ARC model incorporates lifelong learning capabilities by utilizing a task-independent feature extractor. Unlike traditional approaches that require updating the feature extractor for new classes, the ARC model only updates the classifier with fewer parameters.
3) We propose a cross-slicing operation to obtain the multidirectional spatial sequences from HSI. It treats the HSI dataset as spatial sequence data to be fed into an RC model, which is followed by a voting mechanism to obtain the final prediction. The multidirectional input

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and voting mechanism comprehensively considers the spatial–spectral characteristics of the target pixel from four directions, improving accuracy for pixels near the edge.

4) We propose an attention-based leaky function to capture the spatial correlation in HSI data. The leaky is large at the edge of the ROI and decays inside the ROI, which enables the reservoir state to weigh less on the pixels outside the ROI and have a longer memory for pixels inside the ROI. The ARC effectively improves the classification performance.

We demonstrate the effectiveness of the lightweight classification approach on three widely used HSI datasets (Indian Pines, PaviaU, and Salinas), with 93.85%, 98.64%, and 98.89% accuracy, which is superior state-of-the-art lightweight classification models with a comparable training time. We then demonstrate the lifelong learning capabilities of ARC on the three datasets, showcasing a comparable performance to the state-of-the-art lifelong learning models, with a thousand times reduction of parameters to be updated.

The remainder of this article is organized as follows. Section II discusses the related work. Section III describes the proposed ARC, including the framework of the ARC and details about the proposed attention-based leaky function. Section IV presents experiment descriptions. Section V demonstrates that lightweight classification and lifelong learning result on different HSI datasets. Section VI analyzes the experimental results and concludes this article.

II. Related Work

A. HSI Classification Models

In the field of HSI classification, supervised methods have gained considerable attention. Traditional techniques include support vector machines (SVMs) [12], random forest (RF) [13], and multinomial logistic regression (MLR) [14]. Several improvements to these methods have been proposed, but they may suffer from limitations due to handcrafted features and prior information, resulting in reduced classification performance in complex scenarios.

Convolutional neural networks (CNNs) have emerged as a potent tool for HSI classification in recent years, thanks to their ability to learn hierarchical features automatically from input data. CNNs are particularly well-suited for HSI classification tasks because of their local linkage and translational invariance capabilities [15]. They are typically designed to handle spatial and spectral dimensions in HSI classification, allowing them to extract information from both types. This has led to the development of 2-D convolution neural networks (2D-CNNs) and 3-D convolution neural networks (3D-CNNs), which have demonstrated improved performance compared to traditional methods [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]. In addition to CNNs, deep learning models specifically adept at time series classification have been used for HSI classification. These include recurrent neural networks (RNNs) [26], long–short-term memory (LSTM) [27], and convolutional LSTM (ConvLSTM) [28], despite the remarkable performance of deep learning models in HSI classification, they often suffer from over-parameterization, which increases the requirements for computing and storage resources, in both the training and updating of the model.

More recently, there have been inspiring works tailoring the classical models according to the nature of HSI data. The central attention network (CAN) and miniCAN [29] stand out by effectively fusing spectral–spatial features and emphasizing critical information through a scaled dot-product central attention mechanism. The multiarea target attention (MATA) model [30] innovates in multiscale feature extraction, significantly advancing HSI classification by addressing object size variability. The structural optimization transmission network (SOT-Net) [15] leverages cross-attention and dual-mode propagation to optimize multisource data integration for land-cover classification. The spatial-logical aggregation network (SLA-NET) [31] introduces morphological transformations for enhanced fine-grained tree species classification, making notable strides in leveraging morphological differences. Recently, the multilevel domain adaption network (MDA-NET) [32] emerged as a pioneering solution for cross-scene classification, aligning modality-specific distributions and refining classification boundaries through innovative domain adaption techniques. The dual-view spectral and global spatial feature fusion network (DSGSF) [33] enhances HSIC by merging PCA-processed and original HSI features, using spatial and spectral subnetworks for improved classification accuracy. The structure preserved hyper GCN (SPHGCN) [34] introduces a novel approach to modeling complex pixel interactions in HSIC, leveraging hypergraph convolution and a multiple neighborhood reconstruction model for enhanced feature representation. The attention multihop graph and multiscale convolutional fusion network (AMGCFN) [35] innovatively combines multiscale CNNs with multihop GCNs, utilizing attention mechanisms to overcome the limitations of traditional models in HSI classification.

B. Lightweight Classification

The growing scale of DNNs presents challenges in deploying these models on resource-constrained devices, the training cost is hindering the wide deployment of the HSI classification model. To overcome these issues, various model compression techniques have been proposed to reduce network complexity without sacrificing performance [36], [37], [38].

One popular method of model compression is the pruning approach, which eliminates less important parameters based on their impact on feature extraction performance. Other methods include the weight quantization approach, which uses fewer bits to represent network weights to decrease memory consumption; and the knowledge distillation approach, which trains a lightweight target model with knowledge acquired from a high-performance teacher model. MobileNet [36] and SqueezeNet [37] are examples of lightweight architectures that have been specifically designed for efficient deployment on resource-constrained devices. MobileNet utilizes depthwise separable convolutions to reduce computational cost while

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maintaining high accuracy, whereas SqueezeNet employs a compact architecture with fewer parameters without sacrificing performance.

Another critical aspect of HSI classification is feature extraction, and modeling full-image dependencies can greatly improve visual understanding. However, this often leads to significant computational costs. To address this issue, criss-cross operations have been proposed for more efficient feature extraction. Examples of these operations include ACNet [39], which decomposes square-kernel convolution operations into horizontal and vertical linear convolutional operations, and criss-cross attention, which calculates attention within stripe windows in horizontal and vertical directions. While these approaches show promise, their contextual information aggregation performance may be limited by restrictive kernel sizes or configurations. To balance global feature modeling and model efficiency, LiteCCLKNet is proposed to incorporate both large kernels and criss-cross operations, achieving a better tradeoff between performance and computational complexity [40].

C. Lifelong Class Incremental Learning (CIL)

In recent times, lifelong learning has become crucial for HSI classification to deal with the continuous influx of new class data without rebuilding the entire knowledge base [41], [42]. The primary objective of lifelong learning is to adapt to incremental classes with minimal computation overhead. CIL exhibits remarkable robustness in learning knowledge for new classes while retaining previous knowledge, and is commonly employed for HSI classification tasks involving the addition of new classes [43], [44]. The impressive performance of CIL in addressing the challenge of persistent learning for new classes has attracted significant research interest [45].

Current research on CIL typically falls into three main categories. The first, the regularization-based approaches, focus on minimizing the impact of the model on weights when integrating knowledge from new classes. Notable examples include EWC [46] and PathInt [47]. The second category, exemplar-based methods, stores samples from previous tasks as a means of retaining knowledge about earlier-learned classes. Techniques such as iCarL [48] fall into this category. The third type of approach, bias correction, addresses the model's unfavorable leaning toward recently learned tasks, with well-known methods including BiC [49] and WA [50].

In recent years, there has been a growing trend to employ CIL in HSI classification tasks, enabling the continuous learning of new class information. In our previous work, a linear programming incremental classifier (LPILC) was used to tackle the challenge of learning new HSI classes [4]. The geometric spatial–spectral feature integration network with CIL (GS2FIN-CIL) framework aimed at improved adaptation and utilization of these features in processing new classes [51].

D. Echo State Networks (ESNs) and Liquid State Machines (LSMs)

ESNs and LSMs represent pivotal advancements in the domain of RNNs, addressing the challenge of computational intensity often associated with traditional RNN training. Introduced by Jaeger and others, ESNs leverage a reservoir of densely interconnected nodes, each characterized by a nonlinear activation function, to process sequential data efficiently. This reservoir, with connections whose weights are initialized randomly and remain fixed, significantly streamlines the training process by reducing complexity without compromising the network's ability to capture dynamic temporal relationships.

Central to the ESN architecture is its capacity to transform input signals into a high-dimensional, linearly separable feature space through fixed, random reservoir weights, facilitating robust feature extraction and subsequent processing. The framework consists of three key components.

An input layer introduces sequential signals into the system. A dynamic reservoir of randomly connected neurons enables the generation of complex temporal patterns from the input signals. An output layer is responsible for mapping the reservoir's high-dimensional feature representations to the desired outputs.

There are three types of weights in ESNs as follows.

1) The input weight (denoted by W_{in}) governs the connections between the input layer and the reservoir in the ESN. This weight, expressed in an $N \times M$ matrix, determines the value assigned to each connection between input layer nodes and reservoir nodes, with N representing the number of reservoir nodes and M denoting the number of input features. The Input matrix plays a crucial role in transforming input data into a higher dimensional space, allowing the reservoir to capture complex temporal dynamics.
2) The reservoir weight (denoted by W) characterizes the internal connections within the reservoir, specifically the linkages between different reservoir nodes. Such connections help the reservoir elicit rich and diverse dynamical responses to input signals. The reservoir weight matrix has an $N \times N$ size, with initial weights often assigned random values within a particular range, such as between -1 and 1. However, it is common to utilize a normalization method (e.g., spectral radius scaling) to ensure the network operates in the so-called "echo state," resulting in a desirable tradeoff between stability and responsiveness.
3) The output weights are tasked with linking reservoir node states to the desired output. This process employs various linear regression techniques without modifying the input and reservoir weight matrices, leading to a highly efficient training procedure well-suited for modeling time series data and other intricate temporal patterns.

III. ARC for Lightweight-Lifelong HSI Classification

A. ARC Framework

In this section, we introduce the ARC, a lightweight model for lifelong HSI classification. The overall framework is presented in Fig. 2. The data preprocessing steps are shown in Fig. 2(a) and (b), while Fig. 2(c) and (d) depicts the randomly fixed task-independent feature extractor and

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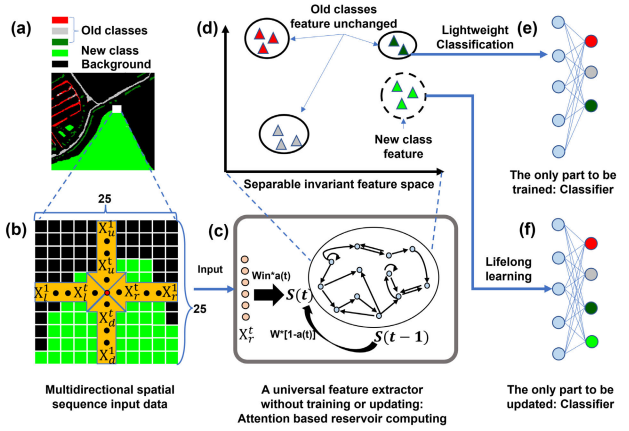


Fig. 2. Framework for ARC HSI classification. (a) HSI includes background and ROI covering old and new classes. (b) Cross-slicing the multidirectional input as spatial sequences for a target pixel. (c) Random fixed reservoir serves as a feature extractor. (d) Mapped separable invariant feature space for old and new class data, where old classes feature remains unchanged when a new class data feature is added. (e) Lightweight classifier for old class classification. (f) Updated classifier for lifelong learning classification of both old and new classes.

the corresponding invariant feature space. The classifier for lightweight classification and lifelong learning classification is shown in Fig. 2(e) and (f), respectively.

1) Cross Slicing for Multidirectional Spatial Sequence: To process a raw HSI, an initial step involves identifying ROI. All the pixels in the ROI need to be finely classified, while pixels outside the ROI are background, as shown in Fig. 2(a). In the context of harnessing HSI data for RC applications, where the inherent model architecture necessitates sequential input, a novel preprocessing approach is imperative. Unlike traditional methodologies that employ a sliding window technique to generate patches of data centered on a target pixel, our method leverages a slicing strategy. This adjustment is crucial due to the unique spatial–temporal dynamics of HSI data, which, unlike linear time series data, embodies multidirectional spatial relationships extending from surrounding pixels toward a target pixel. To aptly capture and preserve these intricate spatial interrelations, we propose the utilization of a cross-slicing technique. This method is distinctively designed to retain the spatial context from multiple directions, thereby enriching the input data with a comprehensive spatial sequence that single-directional slicing or splicing methods fail to achieve.

To preserve the full spectral information, we adopt the cross-slicing method without PCA to obtain the spatial sequence with the full spectral band in Fig. 2(a). For each target spatial pixel in the ROI, the cross-slicing operation obtains four spatial sequences, which come from up X_{u}, down X_{d}, left X_{l}, and right X_{r}, as shown yellow arrays in Fig. 2(b). These four spatial sequences are with the same label and belong to one sample. For the training samples, the four spatial sequences are individually fed into training processes for backpropagation.

For a particular pixel X_T to be classified, four independent inputs X_{u}, X_{d}, X_{l}, and X_{r} will leads to four independent classification results Y_{u}, Y_{d}, Y_{l}, and Y_{r}. A further voting process determines the final classification result Y_T for the pixel X_T.

2) Reservoir Feature Extractor Requiring no Training and Updating: Each spatial sequence serves as independent input data for the ARC model. The reservoir comprises a set of neurons whose value represents the reservoir state. These neurons are interconnected by a random and fixed weight matrix W and are connected to the input layer by another random matrix W_{in}. As the input comes in sequence, the state of the reservoir is updated according to the following equation:

$$S(t) = f(A \times W_{\mathrm{in}} \cdot X^t + (1 - A) \times W \cdot S(t-1)). \quad (1)$$

For example, at step t, the reservoir state $S(t)$ is determined by the previous reservoir state $S(t-1)$ and current spatial pixel X_r^t, as shown in Fig. 2(c) with a certain leaky rate A. Differing from conventional RC, we feed ARC with spatial sequential input. Here, We observe a significant spatial correlation for pixels inside the same ROI. In HSI, the target pixel is typically located in a homogeneous area and is highly related to neighboring pixels in the same ROI, as illustrated in Fig. 2(b). To account for this, we propose an attention mechanism that adjusts a constant leaky rate A to a leaky function $A(t)$ according to the ROI edge information, as will be discussed in Section III-B. Thanks to the chaotic edge nature of RC [52], as introduced in Section II-D, it maps the input to a linearly separable and invariant high-dimension feature space through its fixed random weights, as shown in Fig. 2(d). This property naturally serves as a feature extractor without training, endowing the model with intrinsic lightweight. In addition, the ARC naturally maps new class data to a linearly separate feature space that is nonoverlapping with old class features, as shown in Fig. 2(d). This characteristic makes it a task-independent feature extractor for lifelong learning.

3) Only Part to be Trained or Updated, the Classifier: In the ARC model, only the classifier requires training. The extracted features are used to train a classifier for lightweight classification, as depicted in Fig. 2(e). Upon the arrival of new class data, the corresponding features are extracted using the feature extractor, followed by updating the classifier using old and new class features, as illustrated in Fig. 2(f). This allows the ARC model to achieve lifelong learning.

B. Proposed Attention-Based Leaky Function

In a typical RC, the leaky rate A determines the influence of current input on the final classification result. A higher A gives more weight to the current input and forgets more about previous reservoir states, indicating a higher influence of current input. The leaky rate is typically set to a constant value A in a standard RC range of [0, 1], indicating the same influence of all input pixels. However, in HSI spatial sequence input with significant spatial correlation, the classification result is more influenced by spatial pixels inside the same ROI due to the homogenized nature. Therefore, adjusting the leaky rate according to the significance of each pixel on the target pixel classification is necessary, as shown in the

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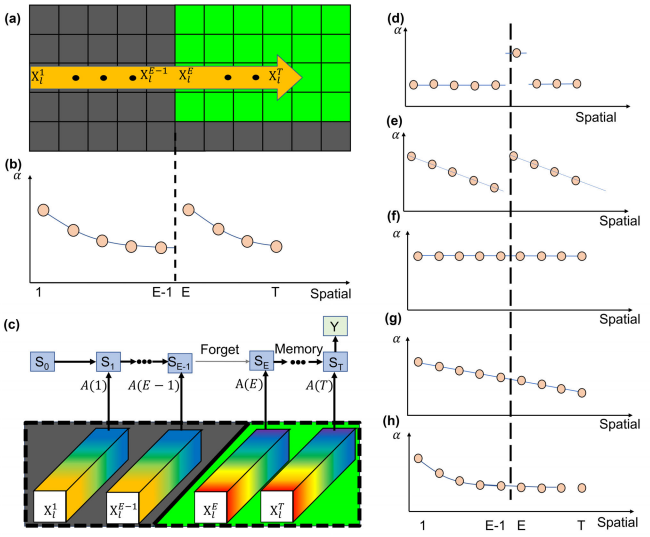


Fig. 3. Illustration of the leaky function of ARC. (a) Input is sliced from the HSI dataset, with T pixels in T steps. (b) Attention exponential decay function. (c) Reservoir pays more attention to pixels in the same homogenized area. (d)–(h) Other kinds of leaky functions for ablation experiments, namely, attention constant function in (d), attention linearly decay attention in (f), constant function in (f), linearly decay function in (g), and exponential decay function in (h).

following equation:

$$S(t) = f(A(t) \times W_{\text{in}} \cdot X^t + (1 - A(t)) \times W \cdot S(t-1)). \tag{2}$$

For instance, the green block in Fig. 3(a) represents an ROI with the same label. Due to the strong spatial correlation, the pixels inside the ROI are more influential on the final classification results, while the pixels outside the ROI have less influence on the classification results. To take such spatial correlation into account, we propose ARC, where the leakage rate is set as a function $A(t)$ of step t that pays more attention to the pixels inside the same ROI as shown in Fig. 3(b). As shown in Fig. 3(c), suppose the pixel X_l^{E-1} is outside the ROI while pixel X_1^E is inside the ROI, the reservoir state $S(E)$ is supposed to be determined more by current input X_1^E instead of the former state $S(E-1)$. This inspires us to the $a(E)$ a higher value to forget the $S(E-1)$. To enable the reservoir state to gain information about pixels inside the ROI, we set the $A(t)$ gradually decay to allow a better memory of the reservoir state.

Our proposed attention-based leaky function shown in Fig. 3(b) is referred to as the attention exponential decay function, which is formulated as follows:

$$A(t) = \begin{cases} a + b \times e^{-t}, & (t < \text{edge}) \\ a + b \times e^{-(t-\text{edge})}, & (t \geq \text{edge}). \end{cases} \tag{3}$$

To investigate the effectiveness of the exponential decay function, a set of attention-based leaky functions are designed as follows.

As shown in Fig. 3(d), attention constant function can be formulated as follows:

$$A(t) = \begin{cases} a, & (t \neq \text{edge}) \\ a + b, & (t = \text{edge}). \end{cases} \tag{4}$$

As shown in Fig. 3(e), attention linearly decay function can be formulated as follows:

$$A(t) = \begin{cases} a - b \times t, & (t < \text{edge}) \\ a + b \times (t - \text{edge}), & (t \geq \text{edge}). \end{cases} \tag{5}$$

To investigate the effectiveness of the attention mechanism, a set of leaky functions without attention to the edge pixels are designed as ablation experiments.

As shown in Fig. 3(f), the constant function is set as follows:

$$A(t) = b. \tag{6}$$

As shown in Fig. 3(g), linearly decay function can be formulated as follows:

$$A(t) = a - b \times t. \tag{7}$$

As shown in Fig. 3(h), exponential decay function can be formulated as follows:

$$A(t) = a + b \times e^{-t}. \tag{8}$$

These functions are tested in the ablation experiment, and the results are given in Section V-D.

C. Innovations of Arc-Over ESN

The ARC model introduces several advancements over the ESN model, specifically tailored to address the complexities of HSI classification. This advancement is manifested through two primary innovations.

1) Multidirectional Spatial Sequence: Unlike the ESN's methodology, which typically generates a single sequence for processing, the ARC model employs a cross-slicing technique. This novel approach facilitates the creation of multidirectional spatial sequences, significantly enhancing the preservation and incorporation of spatial information from various orientations. For each target pixel, multiple spatial sequences corresponding to different directions are generated. A voting mechanism is then applied to integrate the predictive outcomes across these sequences, thereby ensuring a comprehensive consideration of spatial correlation from different directions.

2) Attention-Based Leaky Function: The ARC model introduces an attention mechanism into the leaky function, diverging from the ESN's use of a constant leaky rate. This modification allows the ARC to dynamically adjust its focus at the edge of ROI, capitalizing on the inherent spatial correlations to improve classification performance.

These enhancements not only underscore the ARC model's innovative approach but also its superior suitability for handling the specific challenges presented by HSI classification tasks. The experimental validation of these advancements, as detailed in our ablation study, further demonstrates the ARC model's enhanced performance and its contribution to the field of RC.

IV. Experiment Description

A. Dataset Description

Demonstration of ARC lightweight classification with lifelong learning ability is carried out on three datasets, namely, Indian Pines, PaviaU, and Salinas.

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Indian Pines is an HSI dataset obtained by a single landscape in Indiana with a size of 145 × 145, containing 16 classes with a spatial resolution of 20 m/pixel. The raw Indian Pines dataset includes 220 spectral bands. For research purposes, the remaining 200 bands are selected into ARC after removing the 20 water-related bands. Indian Pines is a challenging dataset for typical HSI classification because of the relatively less training data and relatively large spectral band compared to the PaviaU and the Salinas datasets. Thus, we take 10% data as the training data.

PaviaU is HSI data taken by ROSIS over Pavia University in 2003, which is 610 × 340 in size and contains nine types of objects with a spatial resolution of 1.3 m/pixel. After removing the 12 bands with noise, 103 bands remain labeled, and 6% of them are taken as training data.

Salinas is a HSI dataset taken by AVIRIS in the Salinas Valley, Canada, which contains 16 features with a spatial resolution of 3.7 m/pixel. Twenty noisy bands are removed, leaving 204 bands, the data used in the final experiment. We take 4% of data as the training data.

Specifically, for each target spatial pixel within these datasets, we implement a cross-slicing technique to generate four spatial sequences. Each of these sequences spans a length of 13 spatial pixels, as shown in Fig. 2(b), To keep the training data consistent, we use a 25 × 25 patch as the input for comparison models requiring image-like input.

B. ARC Setting

As the reservoir serves as the feature extractor, we used a high-dimensional reservoir with 1000 reservoir nodes for complex temporal dynamics. The reservoir sparsity and spectral radius are essential hyperparameters to be optimized. In this study, we optimized the sparsity and spectral radius on the PaviaU dataset. The weight matrix W was designed to have 15% connectivity, meaning that only 15% of its elements were nonzero. The optimized spectral radius of W is 0.6. We used the binomial distribution with W_{in} as the number of trials, with a probability of success set at 0.5. For each input spatial sequence, the initial state of the reservoir is set to be all zeros, which prevents the model from being biased by the initial state. Following the acquisition of reservoir states, we employ PCA to identify and select the 75 most representative nodes. These nodes serve as the extracted features, offering a compact yet informative representation of the input data's intrinsic characteristics. For visualization, we further select the top 15 representative nodes. The efficacy of this feature selection method is evidenced by the clear separability of average class features, as illustrated in Fig. 4, which shows the separability of the features extracted from the reservoir and underscores their potential to facilitate accurate classification.

Then, a multilayer perceptron with two hidden layers with 30 nodes in each layer is used as the classifier.

To train the classifier, we use the SGD optimizer with a learning rate of 0.001, and weight decay is set as 0.00001, respectively. The number of epochs is 200, and we use early stopping to prevent overfitting.

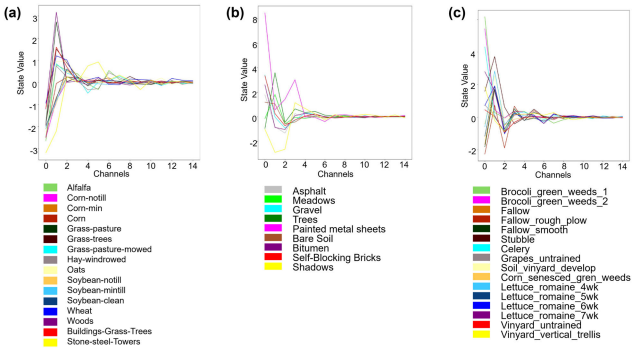


Fig. 4. Average feature of each class for (a) Indian Pines, (b) PaviaU, and (c) Salinas datasets.

TABLE I

NETWORK STRUCTURE OF ORIGINAL MODEL TRAINED ON THE INDIAN PINES DATASET

Layer	Output shape	Weights name	Weights shape	Training
Input	(None, 1000)	W_{in}	(1000, 200)	False
Reservoir	(None, 1000)	W	(1000, 1000)	False
PCA	(None, 75)	-	-	False
Dense1	(None, 30)	W_{out1}	(30, 75)	True
Dense2	(None, 30)	W_{out2}	(30, 30)	True
Dense3	(None, 2)	W_{out3}	(2, 30)	True

C. Experiment Setting

To benchmark the lightweight classification ability, ARC was trained and tested on three HSI datasets. Three representative HSI classification methods, including SVM [12], 2D-CNN [16], 3D-CNN [22], and four state-of-the-art lightweight models, including SqueezeNet [37], MobileNet [36], LiteC-CLKNet [40], and ULite-R2HCN [53] were selected for comparison. The training time of each model was recorded to illustrate the lightweight training process.

To demonstrate lifelong learning ability, ARC was trained on two classes of each dataset, followed by incremental learning on the rest classes one by one. Take the Indian Pines dataset as an example, the ARC was first trained on two classes, with the model structure shown as Table I. A critical aspect of adapting our model to various HSI datasets, each characterized by distinct spectral dimensions, involves the precise configuration of the input weight matrix, W_{in}. Specifically, the dimensionality of W_{in} is aligned with the spectral characteristics of the target HSI dataset. For instance, in the context of the Indian Pines dataset, which comprises 200 spectral bands, coupled with a reservoir of 1000 nodes, the resulting dimensionality of W_{in} is established as 1000 × 200. This configuration ensures seamless integration and processing of HSI datasets across varying spectral dimensions within our model's architecture. The test data of the first two classes were used to evaluate the original model performance on the first two classes' data.

When new class data arrives, the model uses the same W_{in} and W to take the feature of the new class training data. Then, the W_{out1}, W_{out2}, and W_{out3} are updated using features extracted from old classes training data and new class training data, getting a model as shown in Table II. Then the updated model was tested on both the old classes and the new class.

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TABLE II
NETWORK STRUCTURE OF UPDATED MODEL ON INDIAN PINES UPON THE ARRIVAL OF THE FIRST NEW CLASS DATA

Layer	Output shape	Weights name	Weights shape	Updating
Input	(None, 1000)	W_{in}	(1000, 200)	False
Reservoir	(None, 1000)	W	(1000, 1000)	False
PCA	(None, 75)	-	-	False
Dense1	(None, 30)	W_{out1}	(30, 75)	True
Dense2	(None, 30)	W_{out2}	(30, 30)	True
Dense3	(None, 3)	W_{out3}	(3, 30)	True

For comparison, the iCaRL [48], BiC [49], WA [50], CCIL [54], GS2FIN-CIL [51], ECA-MSDWNet [55], and DCPRN [56] were compared with ARC. As these compared lifelong learning model requires updating the task-dependent feature extractor, which is computationally intensive, these models were set to incrementally learn four classes, three classes, and four classes on the Indian Pines, PaviaU, and Salinas datasets, respectively.

All experiments were repeated ten times to get the average and standard deviation of the classification results.

The experiment is conducted on the hardware with 2 GPUs (NVIDIA 1080Ti) and 16 CPUs (Intel Xeon CPU E5-2620 v4 at 2.10 GHz). The software environment is the Ubuntu 16.04 64-bit and deep learning frameworks of Tensorflow.

V. RESULTS AND ANALYSIS

In this section, we evaluate the performance of our proposed lightweight classification model with lifelong learning capabilities, ARC, and compare its performance with other state-of-the-art models.

A. Parametric Study

The attention exponential decay function is the key for ARC to incorporate the spatial correlation in HSI data. The parameters in the function dominate the extent of the reservoir state to forget and memorize the previous information. The smaller a is, the longer the memory of the reservoir state. The larger b is, the stronger the attention to the pixel inside the ROI, which means the reservoir state will forget the information outside the ROI faster. In our exploration to identify optimal parameters for the leaky function $A(t)$, we undertook a comprehensive parametric study across three distinct datasets. It is crucial to note that the function $A(t)$ is governed by a constraint, namely, $a + b <= 1$, to ensure its leaky rate remains between 0 and 1. The results of this study, detailed in Table III, highlight the optimal parameter configuration as $a = 0.1$ and $b = 0.5$ across all investigated datasets. This selection was based on achieving the best balance between adaptability and stability within the model's leaky integration function, thereby optimizing its performance.

TABLE III
OA (%) OF DIFFERENT LEAKY FUNCTION PARAMETER ON INDIAN PINES, PAVIAU, AND SALINAS

Dataset		b=0.1	b=0.5	b=0.9
IP	a=0.1	83.51	**93.85**	84.87
	a=0.5	89.15	87.38	-
	a=0.9	61.94	-	-
PU	a=0.1	90.28	**98.64**	90.26
	a=0.5	97.33	93.27	-
	a=0.9	72.50	-	-
SA	a=0.1	82.91	**98.89**	91.83
	a=0.5	98.07	95.23	-
	a=0.9	79.42	-	-

B. Lightweight Classification Performance

To verify the lightweight classification performance of the proposed approach, we qualitatively and quantitatively evaluate the classification performance by comparing ARC with the abovementioned baselines, including three conventional HSI classification models: SVM, 2D-CNN, and 3D-CNN, as well as four lightweight classification models: SqueezeNet, MobileNet, LiteCCLKNet, and ULite-R2HCN.

1) Results on the Indian Pines Dataset: Table IV shows the classification accuracy obtained by different methods, with the highest value in each row bolded. The value is averaged over 10 runs, the value in brackets is standard deviations. The proposed ARC achieves the highest classification accuracy for the Indian Pines dataset in terms of overall accuracy (OA) and Kappa. The OA of ARC has a 1.28% improvement over the second optimal model, the ULite-R2HCN model. Compared with lightweight models including SqueezeNet, MobileNet, and LiteCCLKNet, the ARC achieves a 27.12%, 22.14%, and 13.60% improvement in OA, respectively. The ARC also achieves an 11.34%, 13.52%, and 3.72% improvement in average accuracy (AA), respectively. The classification accuracy of ARC is the best in seven classes out of 16 classes. Fig. 5 shows the classification maps of the Indian Pines dataset obtained by different methods. Fig. 5(a) shows the false color image of the Indian Pines dataset. Fig. 5(b) shows the ground truth of the Indian Pines dataset. Fig. 5(f)–(h) shows the classification map obtained by SqueezeNet, MobileNet, and LiteCCLKNet models, showing obvious incorrect classification results due to the classification difficulty in the Indian Pines dataset. Fig. 5(i) shows the classification map obtained by the ULite-R2HCN which achieves the second-best OA. Fig. 5(j) shows the classification map obtained by the ARC, showing the best classification result. We can see the ARC works well for small ROIs for the edge pixels for ROIs, this reflects the effectiveness of the attention-based leaky function and the multidirectional input voting mechanism. The Indian Pines dataset, characterized by numerous narrow area ROIs, presents a unique challenge due to the huge amount of edge pixels that often confound standard lightweight classification models. The ARC model's proficiency in this context can be attributed to its attention-based leaky function, which is adept at processing edge pixels. This function likely enhances the model's sensitivity to the distinct spectral signatures of edge pixels, which are critical in accurately classifying fragmented landscapes. The ARC model's robust performance, particularly in such a complex dataset, implies that its attention mechanisms are effectively prioritizing relevant features while mitigating the influence of mixed pixels common at the edges of small ROIs. This capability not only contributes to the model's high accuracy in individual classes but also to its overall classification performance, as evidenced by its substantial OA

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TABLE IV
COMPARISON OF EXPERIMENTAL LIGHTWEIGHT CLASSIFICATION RESULTS ON INDIAN PINES

Class	SVM	2D-CNN	3D-CNN	SqueezeNet	MobileNet	LiteDenseNet	ULite-R2HCN	ARC
Alfala	54.24(12.92)	69.44(5.65)	52.38(10.11)	86.64(4.47)	83.78(1.93)	94.34(1.13)	94.80(1.00)	**95.65(2.47)**
Corn-notill	64.85(5.13)	53.13(10.23)	67.29(7.60)	50.19(7.66)	48.97(9.14)	40.58(11.95)	89.72(2.78)	**92.72(3.01)**
Corn-min	69.63(6.89)	81.13(7.88)	**99.41(0.13)**	59.11(6.19)	57.51(5.25)	70.90(5.86)	95.47(4.98)	90.36(4.27)
Corn	58.33(7.33)	41.35(15.07)	82.89(9.33)	88.84(3.92)	78.32(2.23)	**96.24(0.78)**	94.35(3.48)	83.97(5.99)
Grass/Pasture	81.82(6.21)	87.44(6.75)	88.41(6.55)	72.79(5.73)	69.32(3.32)	78.34(4.15)	**97.82(1.31)**	93.79(5.63)
Grass/Trees	87.07(3.86)	92.18(3.96)	89.77(6.23)	84.61(4.98)	90.37(0.37)	92.99(1.86)	99.10(0.87)	**99.45(0.18)**
Grass/Pasture-mowed	83.33(6.50)	55.00(10.09)	29.17(17.33)	52.17(7.38)	38.14(7.66)	48.99(9.69)	**99.48(0.20)**	53.57(17.52)
Hay-windrowed	96.94(2.71)	92.94(3.57)	100.00(0.00)	77.69(5.25)	95.86(0.14)	94.65(1.00)	**100.00(0.00)**	99.58(0.17)
Oats	28.57(17.52)	76.92(8.47)	11.11(6.25)	68.13(5.92)	36.85(8.85)	60.50(7.68)	**80.00(10.40)**	50.00 (20.92)
Soybeans-notill	55.19(10.38)	79.77(7.69)	95.13(2.10)	59.23(5.97)	66.49(3.51)	91.32(2.10)	92.10(4.35)	**95.68(3.86)**
Soybeans-min	65.31(5.63)	66.33(9.34)	98.63(0.95)	52.96(7.27)	74.12(2.88)	89.86(3.76)	85.61(3.45)	**93.77(3.21)**
Soybean-clean	65.27(5.99)	59.92(12.25)	65.09(7.77)	55.72(6.83)	38.44(7.64)	62.95(6.05)	**94.31(1.26)**	90.39(7.33)
Wheat	85.17(4.27)	99.44(0.21)	97.79(2.43)	97.49(2.45)	94.60(0.40)	99.05(0.24)	98.63(0.10)	**100.00(0.00)**
Woods	91.98(3.01)	80.78(6.44)	97.73(1.39)	93.70(3.11)	93.31(0.69)	88.71(3.78)	98.56(1.02)	**99.53(0.23)**
Building-Grass-Trees-Drives	63.54(4.68)	75.73(8.86)	78.29(8.86)	86.05(4.34)	82.87(1.13)	**94.86(1.23)**	94.19(5.13)	81.35(5.92)
Stone-steel Towers	95.83(2.47)	94.94(2.45)	84.34(9.56)	95.50(2.28)	97.09(0.91)	**98.45(0.29)**	95.23(2.99)	92.47(3.33)
OA	72.13(1.22)	73.14(1.25)	88.85(1.47)	66.73(5.81)	71.71(1.29)	80.25(1.77)	92.57(1.01)	**93.85(1.08)**
AA	71.69(2.81)	75.40(1.98)	77.37(1.69)	73.80(2.07)	71.62(1.38)	81.42(2.22)	**94.33(0.94)**	85.14(1.27)
KAPPA	67.84(0.98)	69.63(1.13)	87.10(0.92)	71.25(1.59)	69.11(1.89)	81.37(1.26)	90.12(1.36)	**92.86(1.47)**

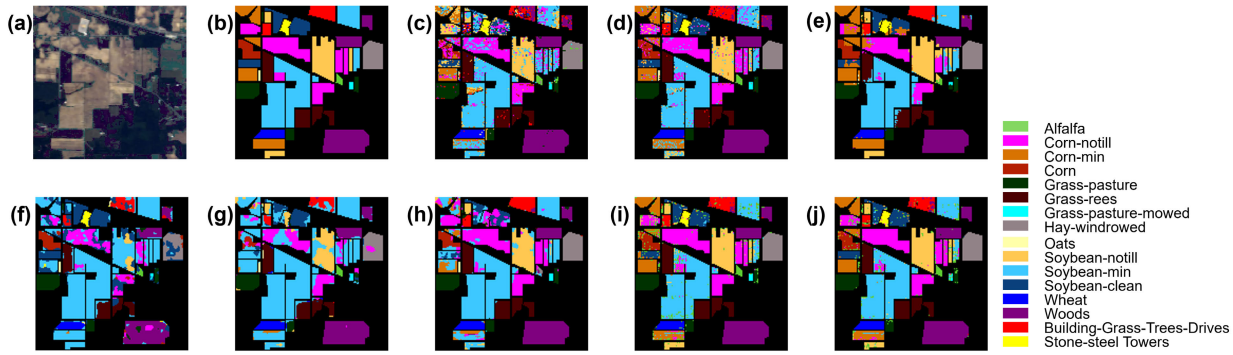


Fig. 5. Classification maps obtained by different methods on the Indian Pines dataset. (a) False color map. (b) Ground truth of Indian Pines data. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) SqueezeNet. (g) MobileNet. (h) LiteDenseNet. (i) ULite-R2HCN. (j) ARC. Class legend is shown on the right.

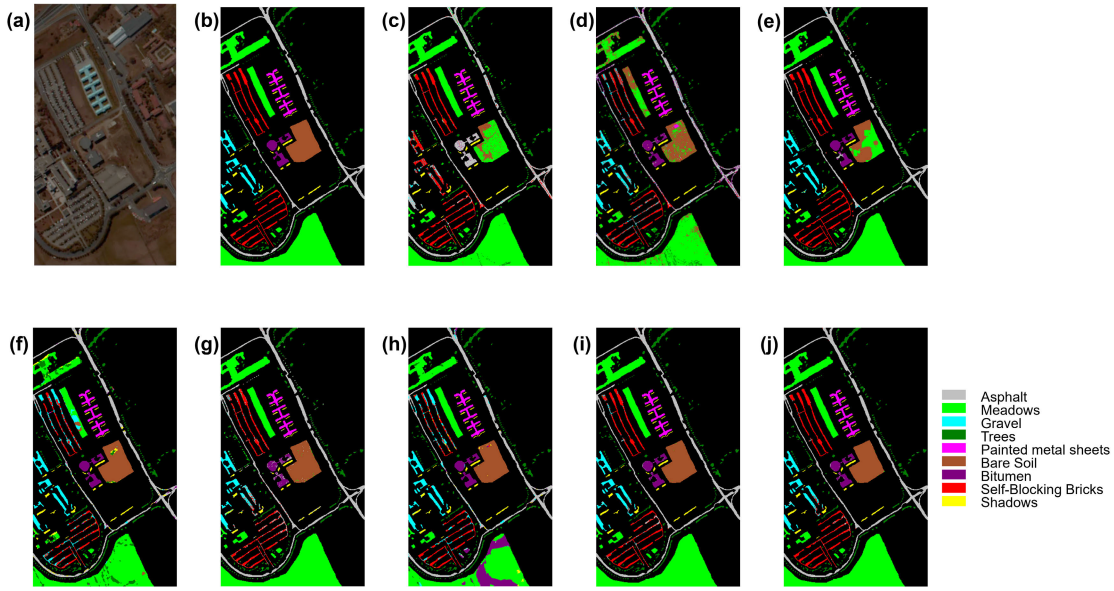


Fig. 6. Classification maps obtained by different methods on the PaviaU dataset. (a) False color map. (b) Ground truth of PaviaU data. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) SqueezeNet. (g) MobileNet. (h) LiteDenseNet. (i) ULite-R2HCN. (j) ARC. Class legend is shown on the right.

and Kappa scores. The effectiveness of ARC in handling the intricacies of the Indian Pines dataset underscores its potential as a specialized tool for areas with fine-grained land-cover distinctions.

2) Results on the PaviaU Dataset: The classification accuracy obtained by different methods on the PaviaU dataset is summarized in Table V. The value is averaged over 10 runs, the value in brackets is standard deviations. The highest

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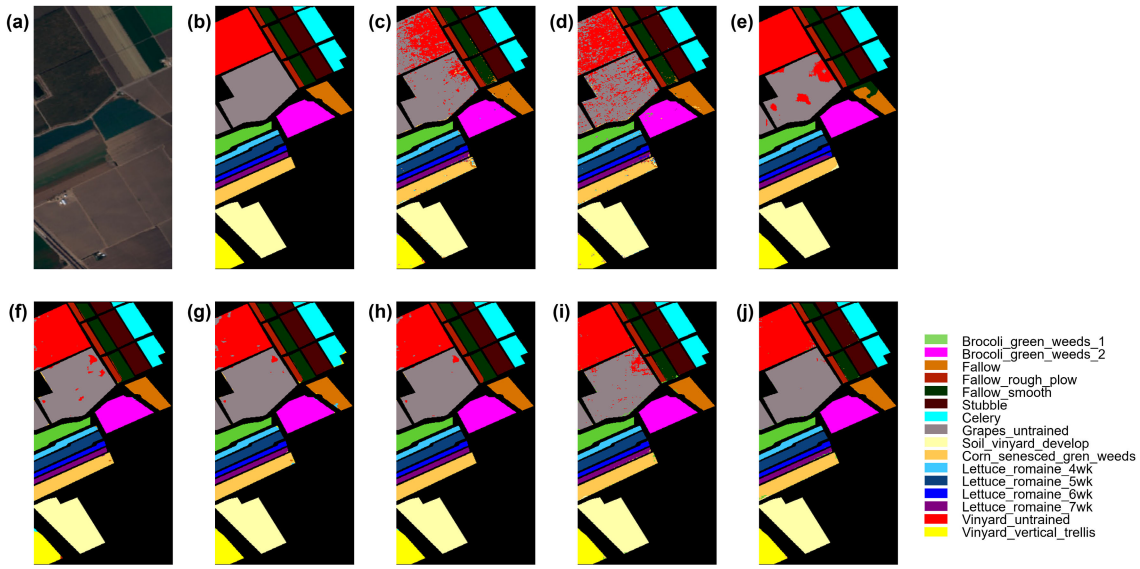


Fig. 7. Classification maps obtained by different methods on the Salinas dataset. (a) False color map. (b) Ground truth of Salinas data. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) SqueezeNet. (g) MobileNet. (h) LiteDenseNet. (i) ULite-R2HCN. (j) ARC. Class legend is shown on the right.

TABLE V

COMPARISON OF EXPERIMENTAL LIGHTWEIGHT CLASSIFICATION RESULTS ON PAVIAU

Class	SVM	2D-CNN	3D-CNN	SqueezeNet	MobileNet	LiteDenseNet	ULite-R2HCN	ARC
Asphalt	75.64(4.01)	83.70(5.86)	97.35(1.48)	92.03 (3.24)	92.53 (4.80)	93.24 (5.16)	96.72(2.32)	**99.50(0.50)**
Meadows	80.18(3.47)	93.35(1.72)	99.19(0.23)	92.64 (4.56)	93.89 (3.87)	78.35 (11.09)	96.08(3.05)	**99.58(0.12)**
Gravel	76.34(3.93)	54.01(11.54)	78.53(8.89)	84.43 (5.73)	90.21 (2.99)	**99.43 (5.27)**	97.64(2.00)	92.75(3.35)
Trees	94.39(1.79)	87.37(3.39)	94.95(2.47)	95.85 (1.00)	96.38 (2.46)	**99.74 (2.47)**	97.98(1.90)	98.86(1.04)
Painted metal sheets	99.29(0.36)	99.92(0.00)	94.04(1.92)	98.90 (1.36)	98.63 (1.07)	**99.85 (0.24)**	98.94(0.73)	99.39(0.61)
Bare Soil	94.30(1.80)	57.45(10.74)	59.84(13.36)	93.17 (3.83)	95.16 (3.39)	98.85 (1.74)	95.59(0.51)	**99.35(0.45)**
Bitumen	51.53(34.05)	82.17(5.06)	84.00(7.00)	87.19 (4.68)	90.98 (5.41)	**99.25 (0.43)**	97.24(2.31)	95.70(4.30)
Self-Blocking Bricks	66.96(6.24)	84.43(4.71)	91.98(4.49)	95.55 (2.50)	94.30 (3.51)	64.83 (15.80)	**97.95(1.05)**	95.13(4.87)
Shadows	**100.00(0.00)**	99.20(0.55)	98.36(1.12)	96.63 (2.38)	93.56 (3.07)	96.94 (2.81)	98.21(0.88)	99.50(0.42)
OA	81.68(1.46)	84.50(1.66)	92.59(1.21)	92.80 (1.01)	93.91 (0.71)	86.21 (2.77)	96.67(1.73)	**98.64(0.76)**
AA	76.35(1.93)	82.40(1.98)	88.69(0.66)	92.93 (0.82)	93.96 (0.73)	92.28 (2.07)	97.37(1.91)	**97.75(0.25)**
KAPPA	72.11(1.84)	79.23(2.14)	90.27(1.14)	90.21 (1.59)	91.02 (1.38)	82.65 (3.74)	95.75(1.37)	**97.37(0.63)**

TABLE VI

COMPARISON OF EXPERIMENTAL LIGHTWEIGHT CLASSIFICATION RESULTS ON SALINAS

Class	SVM	2D-CNN	3D-CNN	SqueezeNet	MobileNet	LiteDenseNet	ULite-R2HCN	ARC
Brocoli_green_weeds_1	**99.95(0.01)**	98.18(1.12)	99.93(0.02)	99.51 (0.58)	99.80 (0.05)	99.86 (0.10)	93.47(1.74)	99.71(0.12)
Brocoli_green_weeds_2	99.55(0.11)	98.99(0.91)	**100.00(0.00)**	99.64 (0.16)	99.76 (0.22)	**100.00 (0.00)**	95.35(2.09)	99.71(0.41)
Fallow	96.03(0.98)	92.86(4.08)	72.58(9.42)	98.80 (0.26)	98.85 (1.23)	98.41 (0.50)	96.96(3.34)	**99.56(0.59)**
Fallow_rough_plow	97.23(0.69)	99.17(0.72)	98.95(0.12)	**99.82 (0.19)**	95.44 (4.39)	98.14 (0.55)	93.56(3.80)	99.57(0.38)
Fallow_smooth	99.32(0.17)	97.58(1.36)	**99.87(0.05)**	96.99 (2.02)	95.89 (2.67)	96.70 (2.63)	95.11(2.40)	98.63(1.22)
Stubble	**99.95(0.01)**	99.94(0.05)	99.91(0.01)	99.99 (0.03)	99.74 (0.42)	99.59 (0.17)	96.15(2.34)	99.61(0.47)
Celery	99.30(0.18)	99.18(0.72)	**100.00(0.00)**	99.20 (0.23)	98.92 (1.11)	99.80 (0.14)	96.67(1.94)	99.35(0.24)
Grapes_untrained	78.04(5.53)	85.54(6.65)	86.92(9.18)	78.09 (3.02)	88.26 (3.73)	89.70 (1.35)	96.98(2.35)	**98.88(1.18)**
Soil_vinyard_develop	98.56(0.36)	99.11(0.76)	**100.00(0.00)**	99.81 (0.11)	99.42 (0.55)	99.09 (0.16)	99.06(0.85)	99.66(0.21)
Corn_senesced_gren_weeds	95.52(1.12)	92.80(4.11)	97.49(1.01)	96.14 (1.68)	96.84 (1.82)	95.78 (2.73)	97.62(1.48)	**98.06(0.74)**
Lettuce_romaine_4wk	94.37(1.41)	94.31(3.80)	**99.89(0.01)**	98.61 (1.22)	99.50 (0.84)	96.00 (2.23)	99.28(0.89)	99.62(0.40)
Lettuce_romaine_5wk	97.57(0.61)	99.95(0.05)	**100.00(0.00)**	**100.00 (0.00)**	99.40 (0.24)	99.93 (0.14)	98.76(0.55)	99.66(0.35)
Lettuce_romaine_6wk	93.53(1.62)	89.02(5.67)	**100.00(0.00)**	98.04 (1.33)	97.59 (1.86)	98.32 (0.26)	98.46(0.18)	99.71(0.17)
Lettuce_romaine_7wk	95.26(1.19)	95.89(2.62)	85.90(5.31)	99.77 (0.35)	**99.95 (0.14)**	98.39 (0.87)	98.47(0.71)	96.07(2.02)
Vinyard_untrained	82.36(4.39)	66.29(16.28)	**100.00(0.00)**	93.42 (2.49)	86.74 (5.29)	85.91 (3.97)	98.76(0.72)	97.18(1.26)
Vinyard_vertical_trellis	97.94(0.51)	97.80(1.99)	**100.00(0.00)**	98.51 (0.90)	99.43 (0.19)	98.49 (1.02)	97.46(2.12)	98.60(1.58)
OA	91.70(0.38)	90.87(0.77)	95.61(0.39)	93.89 (0.91)	94.95 (0.42)	95.09 (0.82)	97.17(1.22)	**98.89(0.26)**
AA	95.28(0.68)	94.17(0.91)	96.34(0.66)	97.27 (0.51)	97.22 (0.22)	97.13 (0.86)	97.01(1.20)	**98.97(0.17)**
KAPPA	90.73(0.32)	89.82(0.57)	95.10(0.90)	93.96(0.95)	94.63 (1.02)	94.27(1.35)	96.89(1.70)	**98.55(0.29)**

number for each class accuracy, OA, AA, and Kappa is bolded. The proposed ARC achieves the highest classification accuracy for the PaviaU dataset in terms of OA, AA, and Kappa. The OA and AA of ARC have a 1.97% and 0.38% improvement over the second optimal model, the ULite-R2HCN model. Compared with SqueezeNet, MobileNet, and LiteCCLKNet, the ARC achieves a 5.84%,4.73%, and 12.43% improvement in OA, respectively. The ARC achieves a 4.82%, 3.79%,

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TABLE VII
TRAINING TIME AND FLOPS PER SAMPLE FOR SVM, 2D-CNN, 3D-CNN, SQUEEZENET, MOBILENET, LITEDENSENET, ULITE-R2HCN, AND ARC

Dataset	SVM	2D-CNN	3D-CNN	SqueezeNet	MobileNet	LiteDenseNet	ULite-R2HCN	ARC
IP	185.98s	229.5s	306.88s	51.78s	55.82s	65.83s	35.82s	56.95s
PU	457.35s	270.38s	358.36s	59.69s	58.67s	69.34s	59.45s	65.61s
SA	518.04s	493.83s	568.05s	86.35s	79.97s	90.66s	71.83s	85.98s
FLOPs per sample / M	3.24	1143.52	1335.05	306.44	30.49	120.71	26.39	36.4

and 5.47% improvement in AA, compared with SquueezeNet, MobileNet, and LiteCCLKNet, respectively. Fig. 6 shows the classification maps of the PaviaU dataset obtained by different methods. Fig. 6(a) shows the false color image of the PaviaU dataset. Fig. 6(j) shows the classification map obtained by the ARC. Fig. 6(b) shows the ground truth of the PaviaU dataset. Fig. 6(i) shows the classification map obtained by the ULite-R2HCN which achieves the second-best OA. Fig. 6(f)–(h) shows the classification map obtained by other lightweight models. We can see the ARC can generate more uniform classification results for the inner pixels of ROIs and better classification results for the edge pixels of ROIs. In the classification of the PaviaU dataset, the ARC model demonstrates exceptional versatility, excelling in both narrow areas like Asphalt and broad regions such as Meadow and Bare Soil. This adaptability suggests that the ARC's feature extractor is finely tuned to manage pixels at the edges as well as within the center of ROIs, capturing the full extent of spatial context with high precision. The model's OA, AA, and impressive Kappa coefficient of 97.37% reflect its superior performance and confirm its balanced classification strength across varied land-cover types. Such a high Kappa value points to ARC's significant agreement between predicted and observed classifications, indicating its effectiveness in dealing with PaviaU's diverse landscape.

3) Results on the Salinas Dataset: Table VI shows the classification results of the Salinas dataset. The value is averaged over 10 runs, the value in brackets is standard deviations. The highest number for each class accuracy, OA, AA, and Kappa is bolded. The proposed ARC achieves the highest classification accuracy for the Salinas dataset in terms of OA, AA, and Kappa. The OA of ARC has a 1.72% improvement over the second optimal model, the ULite-R2HCN model. Compared with SqueezeNet, MobileNet, and LiteCCLKNet, the ARC achieves a 5%, 3.94%, and 3.8% improvement in OA, and a 1.7%, 1.75%, and 1.84% improvement in AA, respectively. The classification map of the Salinas dataset obtained by different methods is shown in Fig. 7. Fig. 7(a) shows the false color image of the Salinas dataset. Fig. 7(b) shows the ground truth of the Salinas dataset. Fig. 7(i) shows the classification map obtained by the ULite-R2HCN which achieves the second-best OA. Fig. 7(f)–(h) shows the classification map obtained by lightweight models. Fig. 7(j) shows the classification map obtained by the ARC. We can see the ARC has more uniform classification results for the inner pixels of ROIs.

4) Computational Complexity Analysis: Table VII summarizes the training time and FLOPs (performed per sample) of the proposed ARC model and comparison methods. The training time of the ARC model is composed of two parts, the one-time forward pass time for the reservoir-based feature extractor and the training time for the classifier. Since the reservoir-based feature extractor is fixed, the one-time forward pass time is only incurred once. The training time for the ARC is comparable to the state-of-the-art lightweight classification models. In terms of computational complexity, the ARC requires 36.4 M FLOPs per sample, which is comparable with MobileNet (30.49 M) and ULite-2HCN (26.39 M), and significantly lower than 2D-CNN and 3D-CNN, showing the lightweight property of ARC.

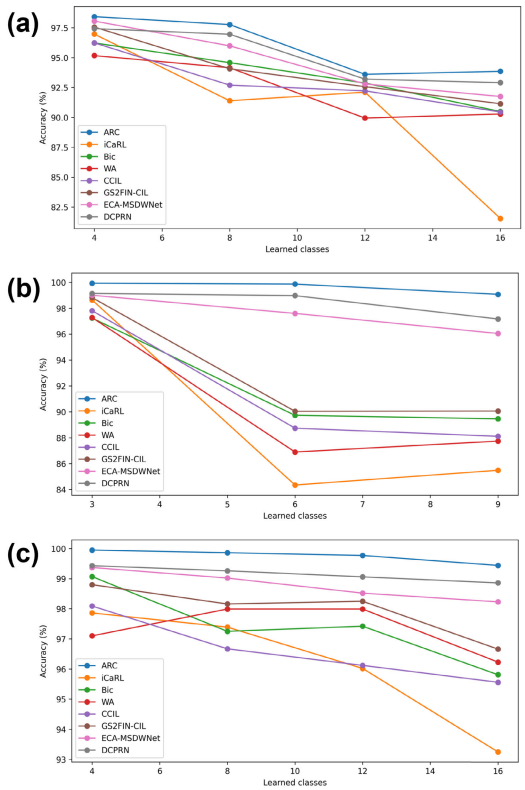


Fig. 8. Comparative lifelong CIL results contains ARC (our method), iCarL [48], BiC [49], WA [50], CCIL [54], GS2FIN-CIL [51], ECA-MSDWNet [55], and DCPRN [56]. OA during the lifelong CIL process for different models on (a) Indian Pines, (b) PaviaU, and (c) Salinas datasets.

C. Lifelong Learning Classification Performance

To verify the lifelong learning effectiveness of the proposed ARC approach, we evaluated it on three HSI datasets in comparison with seven existing incremental learning methods: iCaRL, Bic, WA, CCIL, GS2FIN-CIL, ECA-MSDWNet, and DCPRN. The way our ARC model operates is by initially learning two classes and then incrementally learning new ones. Classification results for class incremental tasks are shown in Tables VIII–X, averaging values over 10 runs, with bracketed values representing standard deviations.

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TABLE VIII
ARC LIFELONG LEARNING ON INDIAN PINES

Class	Number of Learned Classes														
	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Alfala	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	97.83 (1.15)	97.83 (0.22)	97.83 (1.19)	97.83 (2.53)	97.16 (2.04)	97.03 (3.48)	95.65 (5.47)	95.65 (1.27)	97.83 (1.21)	97.82 (1.87)	93.91 (8.69)	95.65 (2.47)
Corn-notill	100.00 (0.00)	100.00 (0.00)	99.72 (0.59)	99.79 (0.55)	99.79 (0.84)	99.81 (0.53)	99.86 (0.55)	99.58 (1.21)	98.11 (0.52)	94.68 (0.27)	94.68 (0.45)	92.37 (0.25)	95.66 (0.57)	92.85 (0.54)	92.72 (3.01)
Corn-min	-	97.46 (1.88)	97.59 (1.85)	96.63 (2.73)	95.66 (2.89)	94.82 (2.25)	94.71 (6.11)	96.99 (3.54)	96.87 (3.53)	91.44 (2.54)	89.40 (0.11)	92.17 (5.21)	92.41 (4.65)	90.12 (1.24)	90.36 (4.27)
Corn	-	-	93.24 (5.48)	91.56 (5.67)	87.34 (7.31)	91.56 (15.67)	89.03 (4.79)	82.71 (10.21)	75.53 (14.58)	76.79 (4.56)	82.71 (6.89)	82.70 (15.43)	83.97 (18.64)	81.86 (18.42)	83.97 (5.99)
Grass/Pasture	-	-	-	98.55 (1.48)	98.34 (1.54)	98.34 (1.98)	97.11 (1.81)	95.65 (3.24)	97.10 (5.87)	95.45 (4.86)	97.32 (0.34)	95.03 (5.81)	94.82 (3.85)	93.40 (2.64)	93.79 (5.63)
Grass/Trees	-	-	-	-	99.73 (0.14)	99.73 (0.33)	99.73 (1.02)	99.45 (0.12)	99.45 (0.24)	99.04 (2.58)	99.32 (2.57)	99.59 (0.35)	99.32 (4.52)	99.60 (0.21)	99.45 (0.18)
Grass/Pasture-mowed	-	-	-	-	-	64.29 (13.58)	82.14 (17.65)	85.71 (20.57)	3.57 (48.64)	78.57 (30.74)	25 (20.58)	7.14 (20.86)	78.57 (10.26)	46.43 (18.59)	53.57 (17.52)
Hay-windrowed	-	-	-	-	-	-	99.79 (0.02)	100.00 (0.00)	99.37 (0.26)	99.58 (2.59)	99.58 (0.69)	99.79 (0.21)	99.79 (0.15)	99.58 (0.01)	99.58 (0.17)
Oats	-	-	-	-	-	-	-	63.75 (13.86)	46.25 (18.50)	62.50 (25.62)	60.00 (29.37)	55.00 (30.21)	37.50 (24.87)	11.25 (11.39)	50.00 (20.92)
Soybeans-notill	-	-	-	-	-	-	-	-	97.63 (2.47)	94.44 (2.56)	95.06 (8.64)	97.22 (6.53)	94.96 (4.67)	95.47 (0.57)	95.68 (3.86)
Soybeans-min	-	-	-	-	-	-	-	-	-	91.36 (1.41)	93.52 (0.52)	92.54 (0.55)	94.50 (1.21)	93.77 (1.56)	93.77 (3.21)
Soybean-clean	-	-	-	-	-	-	-	-	-	-	89.545 (5.86)	86.67 (2.81)	90.56 (5.67)	90.21 (3.25)	90.39 (7.33)
Wheat	-	-	-	-	-	-	-	-	-	-	-	96.58 (2.69)	99.51 (0.05)	99.02 (0.16)	100.00 (0.00)
Woods	-	-	-	-	-	-	-	-	-	-	-	-	99.76 (0.13)	99.76 (0.04)	99.53 (0.23)
Building-Grass	-	-	-	-	-	-	-	-	-	-	-	-	-	86.27 (5.47)	81.35 (5.92)
Stone-steel	-	-	-	-	-	-	-	-	-	-	-	-	-	-	92.47 (3.33)
OA	100.00 (0.00)	99.09 (0.27)	98.43 (0.71)	98.05 (0.78)	97.87 (0.93)	97.78 (0.77)	97.77 (0.72)	97.31 (1.25)	96.48 (1.49)	93.2 (1.24)	93.60 (0.82)	93.11 (0.62)	95.28 (0.77)	93.97 (1.02)	93.85 (1.08)

TABLE IX
ARC LIFELONG LEARNING ON PAVIAU

Class	Number of Learned Classes							
	2	3	4	5	6	7	8	9
Asphalt	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	99.98 (0.34)	99.71 (0.13)	99.50 (0.50)
Meadows	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	99.97 (0.09)	100.00 (0.00)	99.95 (0.08)	99.58 (0.12)
Gravel	-	99.14 (0.24)	99.76 (0.10)	99.38 (0.19)	99.19 (0.77)	99.14 (0.09)	94.81 (2.15)	92.75 (3.35)
Trees	-	-	99.38 (0.06)	99.31 (0.27)	99.28 (0.13)	99.18 (0.07)	99.41 (0.29)	98.86 (1.04)
Painted metal sheets	-	-	-	99.93 (0.13)	99.93 (0.10)	100.00 (0.00)	99.85 (0.25)	99.39 (0.61)
Bare Soil	-	-	-	-	99.90 (0.03)	99.78 (0.01)	99.70 (0.09)	99.35 (0.45)
Bitumen	-	-	-	-	-	98.65 (1.35)	96.92 (3.35)	95.70 (4.30)
Self-Blocking Bricks	-	-	-	-	-	-	94.05 (4.05)	95.13 (4.87)
Shadows	-	-	-	-	-	-	-	99.50 (0.42)
OA	100.00 (0.00)	99.93 (0.04)	99.92 (0.06)	99.89 (0.16)	99.86 (0.18)	99.81 (0.29)	98.97 (0.78)	98.64 (0.76)

On the Indian Pines dataset, ARC begins with an accuracy of 100.00% for the initial two classes. As new classes are introduced, OA decreases to 93.85% when there are 16 classes. Remarkably, the model encounters an inflow of new classes that surpasses the count of existing classes up to seven times, resulting in an accuracy degradation of a mere 6.15%. The accuracy of the two initial classes also decreases as new classes are added. The first two initial classes' accuracy decreases to 95.65% and 92.72% when the number of classes reaches 16. On the PaviaU dataset, the

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TABLE X
ARC LIFELONG LEARNING ON SALINAS

Class	Number of Learned Classes														
	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Brocoli_green_weeds_1	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	99.95 (0.27)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	99.71 (0.12)
Brocoli_green_weeds_2	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	99.92 (0.02)	99.97 (0.24)	100.00 (0.00)	99.97 (0.13)	99.92 (0.28)	99.97 (0.1)	99.71 (0.41)
Fallow	-	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	99.9 (0.09)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	99.49 (0.2)	99.95 (0.16)	99.56 (0.59)
Fallow_rough_plow	-	-	99.71 (0.06)	99.64 (0.13)	99.57 (0.17)	99.71 (0.13)	99.64 (0.3)	99.43 (0.02)	99.35 (0.34)	99.35 (0.21)	99.57 (0.08)	99.78 (0.22)	99.5 (0.17)	99.78 (0.02)	99.57 (0.38)
Fallow_smooth	-	-	-	99.29 (0.06)	99.59 (0.04)	99.07 (0.01)	99.51 (0.05)	99.1 (0.23)	99.44 (0.21)	99.33 (0.0)	99.29 (0.39)	99.33 (0.3)	99.18 (0.14)	99.25 (0.03)	98.63 (1.22)
Stubble	-	-	-	-	99.82 (0.16)	99.87 (0.11)	99.62 (0.32)	99.67 (0.24)	99.77 (0.19)	99.72 (0.32)	99.87 (0.23)	99.52 (0.01)	99.82 (0.24)	99.67 (0.27)	99.61 (0.47)
Celery	-	-	-	-	-	99.94 (0.04)	99.94 (0.16)	99.92 (0.04)	99.89 (0.07)	99.94 (0.27)	99.94 (0.36)	100.00 (0.00)	100.00 (0.00)	99.94 (0.09)	99.35 (0.24)
Grapes_untrained	-	-	-	-	-	-	99.95 (0.14)	99.93 (0.12)	99.86 (0.04)	99.66 (0.18)	99.9 (0.06)	99.97 (0.06)	99.88 (0.2)	99.25 (0.21)	98.88 (1.18)
Soil_vinyard_develop	-	-	-	-	-	-	-	100.00 (0.00)	99.98 (0.2)	100.00 (0.00)	100.00 (0.00)	100.00 (0.00)	99.98 (0.09)	99.98 (0.2)	99.66 (0.21)
Corn_senesced_gren_weeds	-	-	-	-	-	-	-	-	99.08 (0.35)	98.63 (0.15)	98.38 (0.47)	98.17 (0.29)	97.96 (0.02)	98.26 (0.04)	98.06 (0.74)
Lettuce_romaine_4wk	-	-	-	-	-	-	-	-	-	100.00 (0.00)	100.00 (0.00)	99.53 (0.05)	99.25 (0.35)	99.63 (0.07)	99.62 (0.40)
Lettuce_romaine_5wk	-	-	-	-	-	-	-	-	-	-	99.95 (0.15)	100.00 (0.00)	99.95 (0.12)	100.00 (0.00)	99.66 (0.35)
Lettuce_romaine_6wk	-	-	-	-	-	-	-	-	-	-	-	100.00 (0.00)	99.89 (0.36)	100.00 (0.00)	99.71 (0.17)
Lettuce_romaine_7wk	-	-	-	-	-	-	-	-	-	-	-	-	96.07 (0.23)	97.76 (0.11)	96.07 (2.02)
Vinyard_untrained	-	-	-	-	-	-	-	-	-	-	-	-	-	99.5 (0.28)	97.18 (1.26)
Vinyard_vertical_trellis	-	-	-	-	-	-	-	-	-	-	-	-	-	-	98.60 (1.58)
OA	100.00 (0.00)	100.00 (0.00)	99.96 (0.24)	99.8 (0.24)	99.85 (0.04)	99.81 (0.09)	99.86 (0.0)	99.85 (0.11)	99.78 (0.3)	99.7 (0.04)	99.77 (0.06)	99.75 (0.04)	99.59 (0.25)	99.53 (0.14)	98.89 (0.26)

ARC achieves 100.00% accuracy with two initial classes. The OA decreases as new classes are added, and decreases to 98.64% when the number of classes reaches 9. The first two initial classes' accuracy decreases to 99.50% and 99.58% when the number of classes reaches 9. On the Salinas dataset, the ARC achieves 100.00% accuracy with two initial classes. The OA decreases as new classes are added, and decreases to 98.89% when the number of classes reaches 16. The first two initial classes' accuracy decreases to 99.71% and 99.71% when the number of classes reaches 16. The cause of this gradual decrease in classification accuracy with the addition of new classes is due to the constant mapping from input to the reservoir state in the ARC. Each class corresponds to fixed features in the feature space, with additional classes making the space increasingly crowded, complicating classification. However, thanks to the extensive repository of reservoir neurons, the feature space can accommodate these new classes.

The OA of the ARC with five lifelong learning methods is shown in Fig. 8. On the Indian Pines dataset as shown in Fig. 8(a), the ARC model demonstrates robustness in its CIL ability. ARC achieved 98.43% OA for the first four classes, which is the highest among the comparison models. As the number of learned classes increases from 4 to 16, the ARC model shows a gradual decline in accuracy, which is a common trend due to the increasing complexity with more classes. However, ARC decreases only 0.66%, 4.83%, and 4.58% when learning 4, 8, 12, and 16 classes, maintaining one of the highest accuracy rates throughout. The ARC model outperforms other models like iCaRL and BiC, particularly in scenarios with a higher number of classes. On the PaviaU dataset as shown in Fig. 8(b), ARC starts off as the best-performing model when only three classes with 99.93% OA. ARC decreases only 1.07% and 1.29% when learning 6 and 9 classes. The ARC model's accuracy declines more slowly compared to other models as the number of learned classes increases, suggesting the feature extractor effectiveness for all class data. Similarly, on the Salinas dataset as shown in Fig. 8(c), the ARC model shows the best performance at the beginning and decays the most slowly as the number of learned classes increases. The ARC model consistently outperforms most other models, including DCPRN, which shows the second least decline in performance.

Despite only updating classifier output weights (around 4k parameters), ARC outperforms state-of-the-art lifelong learning models which update parameters on the million level. ARC maintains superior performance even without updating the feature extractor as is common with other learning models. This demonstrates that ARC can reduce the parameters to be updated by a thousand times when compared to other models, showcasing its efficiency and potential in lifelong learning tasks.

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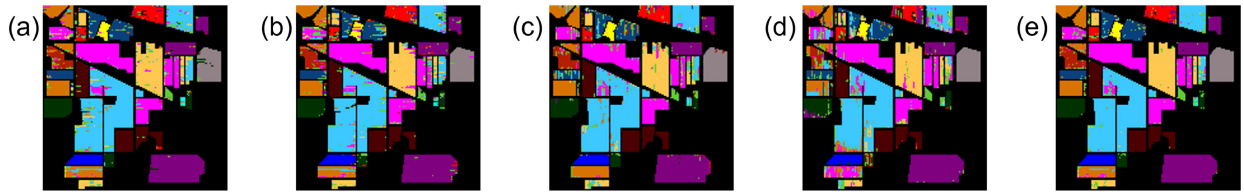


Fig. 9. Classification maps obtained by different directional spatial sequences on the Indian Pines dataset. (a) Left, (b) right, (c) upside, (d) downside, and (e) multidirectional input.

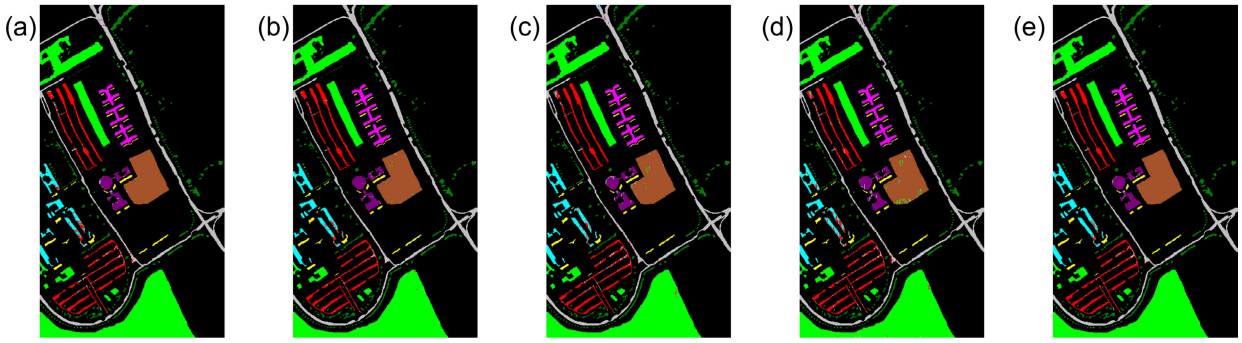


Fig. 10. Classification maps obtained by different directional spatial sequences on the PaviaU dataset. (a) Left, (b) right, (c) upside, (d) downside, and (e) multidirectional input.

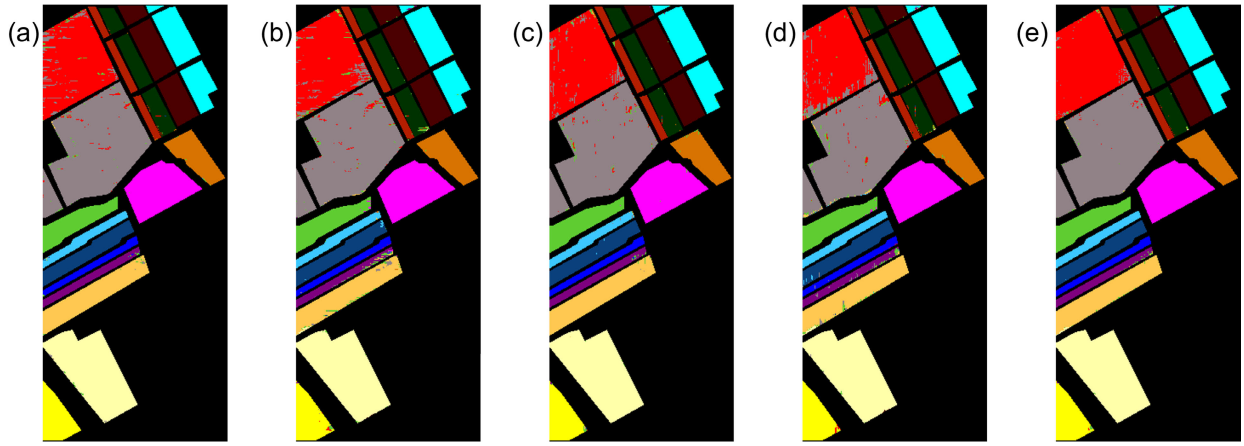


Fig. 11. Classification maps obtained by different directional spatial sequences on the Salinas dataset. (a) Left, (b) right, (c) upside, (d) downside, and (e) multidirectional input.

D. Ablation Experiment

1) Attention-Based Leaky Function: In this section, we demonstrate the effectiveness of the proposed attention-based leaky function through an ablation experiment, where we change the form of the leaky rate and compare the performance of different leaky functions. All types of leaky functions are tested on the three datasets, with OA provided in Table XI, showing the best classification ability of exponentially decaying attention function on all the datasets. As shown in Table XI, the attention exponentially decays function works best for all three datasets. All the function with attention mechanisms performs better than the corresponding function without attention, which reveals the superiority of the attention-based leaky function. It is explainable because the attention mechanism incorporation of the ROI edge information into reservoir state updation enables the reservoir to pay more attention to the pixels inside the ROI. Interestingly, the decayed function performs better than the constant function, which is not observed in RC for time sequence analysis.

2) Multidirectional Input: The data process incorporates a cross-slicing operation that generates four spatial sequences from various directions. This multidirectional input improves the classification accuracy near edges. To validate the efficacy of multidirectional input, we compared its classification results with those from the single-directional input. As shown in Table XII, multidirectional input consistently yielded higher accuracy.

Illustrated in Figs. 9–11 are the classification maps for different directional inputs on three datasets. It's observable that input from the left results in a higher misclassification rate near the left edge [Figs. 9(a), 10(a), and 11(a)], while right-sided input causes more misclassification near the right edge [Figs. 9(b), 10(b), and 11(b)]. Similarly, input from the upside

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TABLE XI
OA (%) OF DIFFERENT LEAKY FUNCTION WITH/WITHOUT ATTENTION

dataset		constant	linearly decay	exponential decay
IP	Without attention	85.15	85.16	83.61
	With attention	90.72	90.51	**93.85**
PU	Without attention	98.28	93.95	97.49
	With attention	98.32	98.26	**98.64**
SA	Without attention	97.96	98.02	98.42
	With attention	98.76	98.78	**98.89**

TABLE XII
OA (%) OF DIFFERENT DIRECTIONAL INPUT AND MULTIDIRECTIONAL INPUT

	Left side	Right side	Upside	Downside	Multidirection
IP	89.96	90.05	89.42	89.75	**93.85**
PU	96.82	97.54	97.19	96.78	**98.89**
SA	97.46	97.05	97.22	96.95	**98.42**

and downside led to increased misclassification near their respective edges. However, implementing a multidirectional input approach, backed by a voting mechanism, effectively decreases misclassification near all edges [Figs. 9(e), 10(e), and 11(e)], consequently enhancing the overall classification performance.

VI. CONCLUSION

The rapid advancements in HSI technology, coupled with the expansion in application scenarios, continually demand lightweight HSI classification models with lifelong learning capabilities. This article introduces the ARC model as a comprehensive solution for these challenges.

The ARC model employs a cross-slicing operation to extract multidirectional input data, treating the HSI dataset as spatial sequence data. This data is subsequently processed within an RC model, with a voting mechanism synthesizing the final prediction. In addition, the integration of an attention-based leaky function allows the ARC model to effectively capture spatial correlation in HSI data, particularly enhancing classification accuracy near edges.

In tests on three well-known HSI datasets—Indian Pines, PaviaU, and Salinas, our proposed ARC model demonstrates impressive accuracy rates of 93.85%, 98.64%, and 98.89%, respectively. These results testify to ARC's superior performance compared to conventional lightweight classification models while maintaining comparable training times. Furthermore, ARC models manifest great potential for lifelong learning. Notably, as new classes increase from two to the total number of classes, the model only experiences minor accuracy decreases of 6.15%, 1.36%, and 1.11% for Indian Pines, PaviaU, and Salinas, respectively. Compared with state-of-the-art lifelong learning models, ARC significantly reduces parameters to be updated a thousand times.

These results underscore the ARC model's ability to deliver lightweight classification while preserving lifelong learning capabilities, thus providing excellent classification performance. Looking forward, we aim to explore the mechanism of the ARC model for spatial sequence data to further enhance feature extraction capabilities and improve classification performance.

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Anran Yuan (Member, IEEE) received the master's degree in electronic science and technology from Xidian University, Xi'an, China, in 2021. She is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Macau University of Science and Technology, Macau, China.

Her research interests include early warning systems for natural hazards, image processing, and artificial intelligence.



Dingchen Wang received the B.S. and M.Eng. degrees from Xi'an Jiaotong University, Xi'an, China, in 2018 and 2021, respectively. He is currently pursuing the Ph.D. degree with the Department of Electrical and Electronic Engineering, The University of Hong Kong, Hong Kong.

His research interests include novel AI computing hardware and material algorithm co-design.



Jing Bai (Senior Member, IEEE) received the B.S. degree in electronic and information engineering from Zhengzhou University, Zhengzhou, China, in 2004, and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2009.

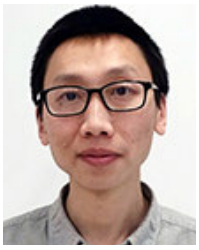
She is currently a Professor with Xidian University. Her research interests include image processing, machine learning, and intelligent information processing.

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Zhu Xiao (Senior Member, IEEE) received the M.S. and Ph.D. degrees in communication and information systems from Xidian University, Xi'an, China, in 2007 and 2009, respectively.

From 2010 to 2012, he was a Research Fellow with the Department of Computer Science and Technology, University of Bedfordshire, Luton, U.K. He is currently a Full Professor with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China. His research interests include mobile computing, image processing, and artificial intelligence.



Zhongrui Wang (Member, IEEE) received the B.Eng. degree (Hons.) and the Ph.D. degree from Nanyang Technological University, Singapore, in 2009 and 2013, respectively. He is an Assistant Professor with the Department of Electrical and Electronic Engineering, The University of Hong Kong (HKU), Hong Kong. He did his Post-Doctoral Research with The University of Massachusetts Amherst, Amherst, MA, USA. His research interest lies in emerging memory based neuromorphic computing and machine learning.



Jianqing Li (Senior Member, IEEE) received the Ph.D. degree from the Beijing University of Posts and Telecommunications, Beijing, China, in April 1999.

From 2000 to 2002, he was a Visiting Professor at Information and Communications University, Daejeon, South Korea. From 2002 to 2004, he was a Research Fellow with Nanyang Technological University, Singapore. He joined the Macau University of Science and Technology, Macau, China, in August 2004, where he is currently a Professor. His research interests are computer network architecture and artificial intelligence.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an. He is currently the Director with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, International Research Center of Intelligent Perception and Computation, Xi'an. His research interests include intelligent information processing, image processing, machine learning, and pattern recognition.

Prof. Jiao is a member of the IEEE Xi'an Section Execution Committee, the President of the Computational Intelligence Chapter, the IEEE Xi'an Section, and the Institution of Engineering and Technology (IET) Xi'an Network, the Chairperson of the Awards and Recognition Committee, the Vice Board Chairperson of the Chinese Association of Artificial Intelligence, a Councilor of the Chinese Institute of Electronics, a Committee Member of the Chinese Committee of Neural Networks, and an Expert of the Academic Degrees Committee of the State Council.

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